UNIVERSITAS GADJAH MADA

Hybrid Handcrafted and CNN Feature Approach

Skin Lesion Classification for Monkeypox Detection

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Outline

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Dataset & Preprocessing
Methodology

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Conclusion

Background

Monkeypox (MPX) is a zoonotic disease caused by the Monkeypox virus.

- Endemic to Central and West Africa
- 2022–2023 global outbreak highlighted urgent need for rapid diagnostics
- Visual inspection is the primary diagnostic method
- Easily confused with chickenpox, measles, and other rashes

Challenge: Automated classification needed for timely diagnosis.

Sample skin lesion images

Problem Statement & Objectives

Problem Statement

Skin lesions from Monkeypox are visually similar to other conditions, making manual diagnosis error-prone and time-consuming.

Research Objectives

1. Robust lesion segmentation from skin background
2. Handcrafted features (color, texture, shape, edges, frequency)
3. Transfer-learned CNN embeddings (ResNet18)
4. Compare handcrafted, CNN, and hybrid representations
5. Evaluate classifiers for binary MPX vs. non-MPX

Dataset Description

Source: Monkeypox Skin Lesion Dataset (Kaggle)

- 228 skin lesion images (224×224 RGB)
- **MPX:** 102 images (45%)
- **Non-MPX:** 126 images (55%)

Split: Stratified 70/30

- Training: 160 images
- Testing: 68 images

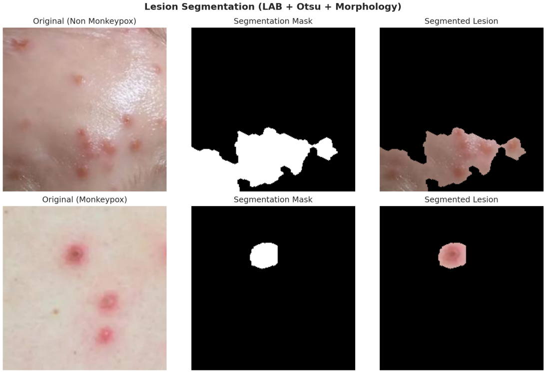
Dataset sample images

Lesion Segmentation Pipeline

Method: LAB + CLAHE + Otsu + Morphology

1. BGR $\rightarrow$ LAB conversion
2. CLAHE on L^* and a^* channels
3. Combine L^* + inverted a^* (0.3, 0.7)
4. Otsu thresholding
5. Morphological closing + opening
6. Keep largest connected component

Result: Clean binary mask isolating the lesion.



Segmentation results

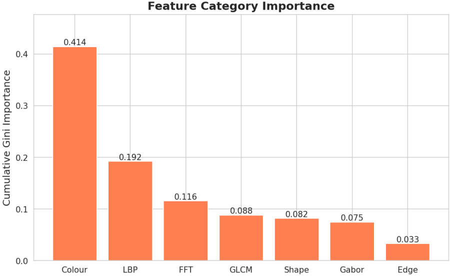
Handcrafted Feature Extraction

Seven categories, ∼200 raw features

- **Color** (84): Moments, HSV hist., percentiles in BGR/HSV/LAB
- **GLCM** (10): Contrast, dissimilarity, homogeneity, energy, correlation
- **LBP** (18): Uniform patterns, r=2, P=16
- **Gabor** (12): 3 orientations × 2 scales
- **Shape** (12): Hu moments + geometric properties
- **Edge** (3): Canny density + Sobel gradient
- **FFT** (16): Radial power spectrum rings

After variance filtering: 155 features

Feature importance (by category)



Cumulative Gini importance

CNN Feature Extraction

Transfer Learning with ResNet18

- Pre-trained on ImageNet (1.2M images)
- Remove final FC layer
- Extract 512-dim embedding
- Inference mode (no gradients)

Preprocessing

- Resize to 224×224
- ImageNet normalization
- $\mu = [0.485, 0.456, 0.406]$
- $\sigma = [0.229, 0.224, 0.225]$

Feature Representations

1. **Handcrafted** (155 dims)
2. **CNN** (512 dims)
3. **Hybrid (full)** (667 dims)
4. **Hybrid (sel.)** (200 dims, MI)

All features Z-score normalized before classification.

Classification Algorithms

Classifier	Hyperparameters
SVM (RBF)	$C = 10$, $\gamma =$scale
Random Forest	200 trees, depth=10
XGBoost	200 trees, depth=6
Logistic Reg.	$C = 1.0$, max iter=5000

- 5-fold stratified CV on training set
- Fixed hyperparameters across all sets
- Independent test set evaluation

Classification Results

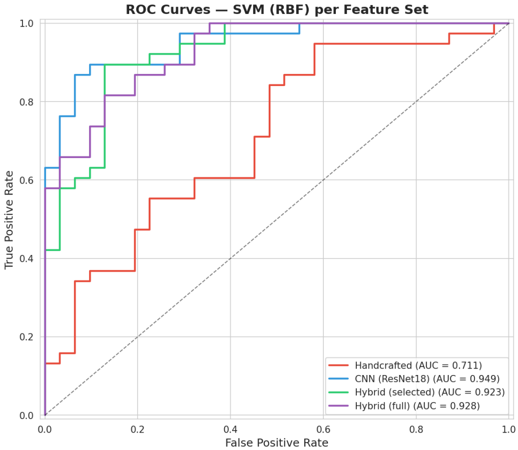
Feature Set	Classifier	Acc.	F1	Test AUC	CV AUC
CNN (ResNet18)	SVM (RBF)	**0.855**	**0.853**	**0.949**	0.891
Hybrid (full)	SVM (RBF)	0.826	0.825	0.928	0.915
Hybrid (sel.)	SVM (RBF)	0.841	0.840	0.923	0.916
CNN (ResNet18)	Random Forest	0.812	0.806	0.911	0.821
CNN (ResNet18)	Logistic Reg.	0.826	0.823	0.887	0.847
Handcrafted	XGBoost	0.754	0.752	0.840	0.768

Key Findings

- CNN-only dominates all baselines
- AUC = 0.949 (near-perfect)
- Hybrid competitive but not superior
- Handcrafted lags significantly

Best: CNN-only + SVM (RBF) → **85.5% accuracy, AUC = 0.949**

ROC Curves



ROC curves for SVM (RBF)

Observations

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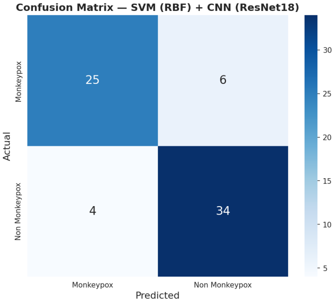
- AUC = 0.949 (near-perfect)
- Hybrid competitive but not superior
- Handcrafted lags significantly

Confusion Matrix & Classification Report



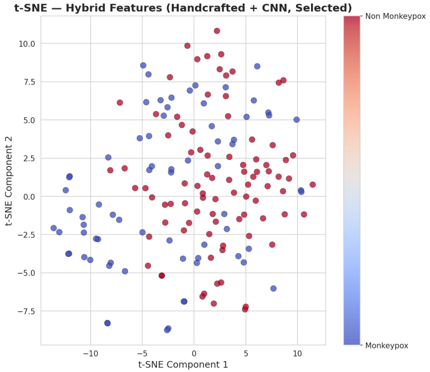
Confusion matrix

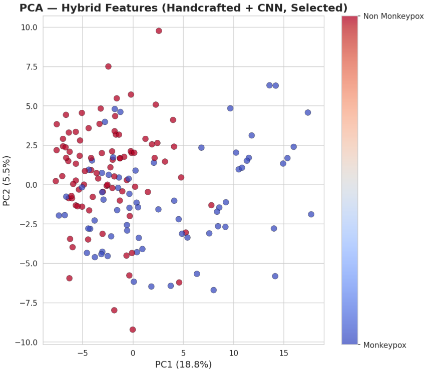
Classification Report (SVM + CNN-only)

Class	Precision	Recall	F1
Monkeypox	0.86	0.81	0.83
Non-Monkeypox	0.85	0.89	0.87
Macro Avg	0.86	0.85	0.85

- 81% MPX recall, 89% non-MPX recall
- Balanced false positives / false negatives
- Overall macro F1 = 0.85

Feature Space Visualization



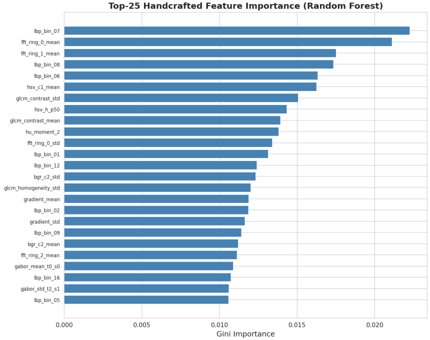


t-SNE (hybrid selected)

PCA (hybrid selected)

Moderate separability; overlap consistent with challenging visual diagnosis

Feature Importance Analysis



Top-25 handcrafted features (RF)

Key Findings

- **Color features** most important (HSV/LAB)
- **GLCM texture** second most discriminative
- Gabor and edge features also contribute
- Aligns with dermatology:
  - MPX: red-purple hues
  - Umbilicated vesicle textures

Discussion

CNN Features

- ResNet18 highly effective **without fine-tuning**
- Captures shape, color, texture patterns
- Best: AUC = 0.949

Handcrafted Features

- Moderate (best AUC = 0.840)
- Color and GLCM most important
- Interpretable and lightweight

Hybrid Approach

- Did **not** surpass CNN-only baseline
- 228 samples: 667 dims overfits
- Selection (200 dims) helps but still below CNN
- May win with larger dataset

Classifier Choice

SVM (RBF) consistently best across all feature sets – non-linear boundary in high-dim space.

Conclusion & Future Work

Summary

1. LAB-based lesion segmentation
2. 7-category handcrafted features (155 dims)
3. ResNet18 CNN embeddings (512 dims)
4. Systematic comparison across 4 classifiers
5. **Best: 85.5% accuracy, AUC = 0.949**

Future Work

- Larger multi-center dataset
- Attention mechanisms for highlighting
- Lightweight mobile deployment
- Multi-CNN ensemble methods

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Thank You

Questions?

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